

A REPORT ON  
Simulating a Minimal Space Interferometric  
Configuration for Low Frequency Radio  
Observations

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## Abstract

We have explored a novel space interferometer system consisting of minimal number (three) of satellites, orbiting the earth in Low Earth Orbits (LEO). Such a system would enable us to observe the radio sky in frequencies under 10 MHz, which is not possible by earth based telescopes, due to the waves being blocked by the earth's atmosphere. This system would also give us huge baselines, greater than the diameter of the earth, which would give us very good resolution for observation. The aim is to obtain the image of the entire radio sky for such low frequencies and spot the differences when compared to other widely accepted radio sky maps. This system would also put to test the limits of a fully space based radio interferometer system and would reveal a whole new kind of data.

# 1 Objectives

The objective of this project was to develop a novel system for performing high quality low frequency radio observations. Radio waves having wavelengths above 30m are blocked by the earth's atmosphere, mainly due to the ionosphere which reflects back the radio waves, thus, not allowing them to reach the surface of the earth. This prevents us from making observations of radio waves with wavelengths above 30m. Therefore, this project was proposed so as to design a system consisting of minimum number of telescopes aboard low earth orbit satellites. We would need multiple satellites because it is not possible to cover the entire sky with just one telescope. Also, our main aim is to observe radio waves with large wavelengths and since the resolution is directly proportional to the wavelength, we would be getting very poor resolution with just one small telescope. So, multiple satellites would be used to make interferometric observations which would increase the resolution several thousand times due to the huge distances between any two satellites at any time. To achieve this primary objective, we were needed to accomplish the following secondary objectives:-

1. Understanding and learning all the concepts involved:-
  - (a) Low frequency radio observations
  - (b) Low earth orbit satellites
  - (c) Very long baseline interferometry
  - (d) Concept of visibility and sky brightness
  - (e) Concept of u-v sampling
  - (f) Learning Python and VPython and all the necessary libraries
2. Deciding upon the minimum number of satellites logically for initial testing
3. Using Python to simulate a realistic system of satellites revolving around the earth, which in turn, is revolving around the sun:-
  - (a) Decide upon the forces acting on the satellites due to the sun and the earth and the forces acting on earth due to the sun
  - (b) Decide upon the time period, and thus, the distances from the earth and orbital velocities of the satellites
  - (c) Decide upon the axes of revolution for the satellites and also the earth, accounting for its axial tilt
  - (d) Make relevant assumptions for the system to work
4. Simulating a source and then observing the source with the satellites:-
  - (a) Source positions are given in RA and Dec coordinates and so, to convert them into relevant coordinates
  - (b) Deciding upon the beam width of the satellites so that the source is visible only when it is within the satellites' beam
  - (c) Deciding upon the reference starting direction and origin of the source

5. Checking the coverage for each source and varying the defined parameters so as to get maximum coverage:-
  - (a) Varying the orbital period of each satellite in different combinations to find the best coverage for all possible source directions
  - (b) Varying the initial phase of each satellite in different combinations to find the best coverage for all possible source directions
6. Quantitatively measuring the coverage for all the three ultimate source directions coarse gridding of the u-v plane and then finding the percentage coverage
7. Correlating the measurements of the voltages from each satellite with each other using VLBI techniques to measure visibilities which when put together are Fourier Transformed to form the final sky map at low radio frequencies
8. Comparing with accepted radio sky maps to spot observational differences

## 2 Introduction

Radio astronomy is a subfield of astronomy that studies the universe in the radio spectrum. To fulfill this purpose, radio astronomy takes multidisciplinary contributions from physics, electronics and computer sciences alike. Domains of antenna design, signal processing, filter design, fiber optics, hardware acceleration, computer modeling etc are as important as astronomy.

A few advantages of observing in the radio frequencies include the fact that most of the higher frequency radio waves are completely allowed to pass through by the Earth's atmosphere, thus enabling ground based observations. It is also possible to observe objects or phenomena that are difficult or impossible to detect in other wavelengths. Radio emissions can also be used for quantitative physical diagnostics of object parameters. But radio waves having wavelengths greater than 30m are blocked completely by the atmosphere, rendering it impossible to make any observations of the radio sky at those wavelengths using ground based telescopes. The only possible solution to make such low frequency studies is to use space based radio telescopes.

To properly observe such large wavelengths with sufficient resolution, the need of telescopes with very large aperture arises. But it becomes impractical to make such huge telescopes and send them to space. Another solution to attain such high resolution is to use the technique of very long baseline interferometry, which uses multiple small aperture telescopes and generate a unified signal by accounting for the delays. The technique of interferometry is a boon for radio astronomers as it enables them to achieve high angular resolution, as the resolving power of an interferometer is set by the distance between its components and not by the size of its components.

This project aims to simulate a space interferometer configuration having a minimum number of low earth orbit satellites so as to be able to provide maximum coverage of the radio sky for frequencies below 10 MHz. These satellites would be able to provide baselines of several thousand kilometers, thus greatly improving the resolution of the observations. We would be using telescopes with



a very wide field of view so as to be able to view the maximum amount of sky, though this would result in a loss of sensitivity. Also, the satellites would be having different orbital periods and would form different baselines at different point of time. This would provide better coverage.

## 2.1 Low Frequency Observations

The radio wavelengths in the electromagnetic spectrum range from 1 millimeter to 100 kilometers. A huge part of this range is allowed to pass through by the earth's atmosphere, enabling us to make ground based radio observations. But for wavelengths greater than 30 meters, they are completely blocked by the atmosphere, making it impossible to observe those waves from earth. Refer to th a linear filter that has a transfer function controlled by variable parameters and a means to adjust those parameters according to an optimization algorithm. Refer to Figure 1 for further details.

In order to make such low frequency observations, we need to have space based observatories. Such an attempt to have a fully space based radio telescope for low frequency observation has not been done till now. My project is to simulate such a space based interferometer using minimum number of satellites.

We are using interferometers because to get sufficiently good resolution for such large wavelengths, it is impractical to build a telescope in space with such huge apertures. interferometry gives us the option to use multiple smaller aperture telescopes to synthesize a larger aperture, equal to the baseline between two telescopes, which in our case, would be thousands of kilometers.

We would be using VLBI techniques to achieve greater resolution, at the expense of collecting area.

$$\theta_{arcsec} \approx 2 \frac{\lambda_{cm}}{D_{km}}$$

where,  $\theta_{arcsec}$  is the resolution of the system, expressed in arcsec,  $\lambda_{cm}$  is the wavelength in cm and  $D_{km}$  is the aperture or baseline of the system.

## 2.2 Low Earth Orbit Satellites

A low Earth orbit (LEO) is an orbit around Earth with an altitude of 2000 km or less, and with an orbital period of between about 84 and 127 minutes. But objects below approximately 160 km will experience very rapid orbital decay and altitude loss due to atmospheric drag, so we usually do not have low earth orbits below 300 km and hence, LEO essentially means an orbit ranging from 300 km to 2000 km. Refer to Figure 2 for further details.

The mean orbital velocity needed to maintain a stable low Earth orbit is about 7.8 km/s, and it reduces with increased orbital altitude. The orbital velocity can be calculated using formula for circular orbit, as LEOs are almost circular in most cases.

$$\text{Orbital velocity, } v = \sqrt{\frac{GM_e}{R}}$$

where,  $G$  is the universal gravitational constant,  $M_e$  is the mass of the earth which is equal to  $5.982 * 10^{24}$  kg and  $R$  is the orbital radius of the satellite.

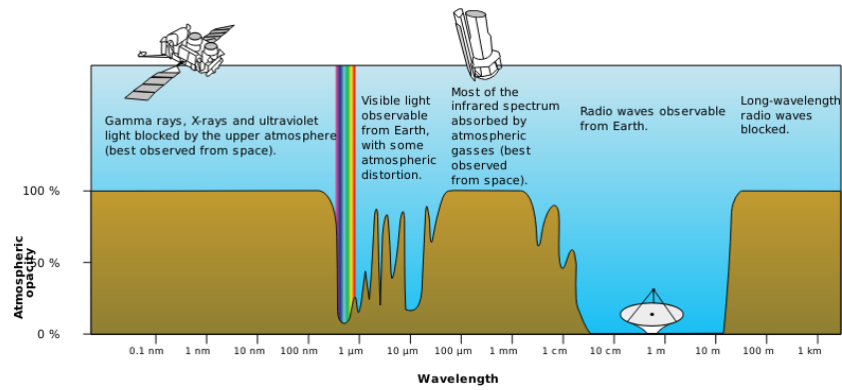


Figure 1: This shows the whole electromagnetic spectrum and the atmospheric opacity for different wavelengths. We can see that for radio waves under 30 m, the opacity is 0% while for wavelengths greater than 30 m, the opacity increases to a 100%, making it impossible to make ground based observations for these wavelengths. (Source: <http://gisgeography.com/electromagnetic-radiation/>)

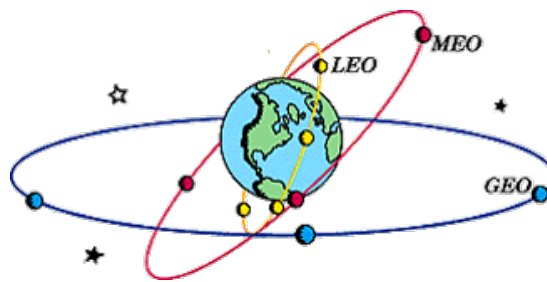


Figure 2: This shows a rough representation (not to scale) of Low Earth Orbit (LEO), Medium Earth Orbit (MEO) and Geostationary Earth Orbit (GEO) satellites. (Source: <http://abyss.uoregon.edu/~js/space/lectures/lec05.html/>)

### 2.3 Very Long Baseline Interferometry (VLBI)

Very-long-baseline interferometry (VLBI) is a type of astronomical interferometry, that is most commonly used in radio astronomy. In VLBI, a signal from an astronomical radio source is collected using multiple radio telescopes placed some distance apart. Next, the distance between the radio telescopes is calculated using the time difference between the arrivals of the radio signal at different telescopes. This time difference is usually very small and is caused as the telescopes are located some finite distance apart. This allows observations of an object that are made simultaneously by many radio telescopes to be combined, thus creating a telescope with a size equal to the maximum separation between the telescopes, which is known as baseline.

Data received at each antenna in the array include arrival times from a local atomic clock which is necessary for correlation of the signals at a later stage. The data from different telescopes are correlated to produce the resulting image. The resolution achievable using interferometry is proportional to the observing frequency. The VLBI technique enables the distance between telescopes to be much greater than that possible with conventional interferometry, which requires antennas to be physically connected by coaxial cable, waveguide, optical fiber, or other type of transmission line. Refer to Figure 3 for further details.

In case of VLBI, the diameter of the single dish radio telescope is replaced by the baseline, which is the distance between any two telescopes of the VLBI array. It is very important to measure the arrival time of the source radiation at each telescope to be able to properly correlate the signals as the telescopes are not physically connected.

VLBI is sometimes also known as aperture synthesis, as observations of the same source made over a certain period of time can also account for movement of the telescopes forming newer baselines, thus, synthesizing the aperture of an equivalent telescope having a diameter equal to the maximum baseline of the array.

### 2.4 Visibility

Visibility, in radio astronomy is defined as the two dimensional Fourier transform of the sky brightness distribution.  $V(u,v)$ , the complex visibility function, is the 2D Fourier transform of  $T(l,m)$ , the sky brightness distribution. Refer to Figure 4 for further details.

$$V(u, v) = \iint T(l, m) e^{-i2\pi(ul+vm)} dl dm$$

Inversely,

$$T(l, m) = \iint V(u, v) e^{i2\pi(ul+vm)} du dv$$

where,  $l, m$  are E-W and N-S angles measured in the tangent plane in radians and  $u, v$  are E-W and N-S spatial frequencies measured in wavelengths.

Some properties of the visibility function and visibility are:-

1. Each  $V(u, v)$  contains information on  $T(l, m)$  everywhere, not just at a  $(l, m)$  coordinate, or within a particular sub-region.

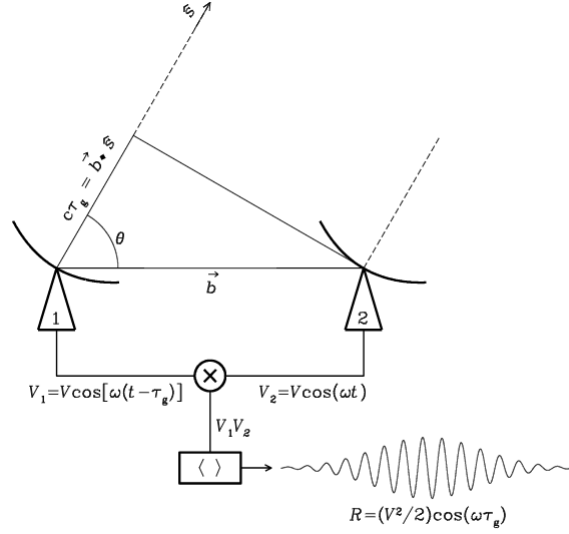


Figure 3: This shows a diagrammatic representation of a normal 2-element interferometer and the steps that are needed in order to get the final image. (Source: <http://www.cv.nrao.edu/course/ast534/>)

2.  $V(u=0, v=0)$  is the integral of  $T(l, m) dl dm$  = total flux density.
3. Each  $V(u, v)$  is a complex quantity - expressed as (*real, imaginary*) or (*amplitude, phase*) but in radio astronomy applications, we usually use only the (*amplitude, phase*) representation. Refer to Figure 5 for further details.
4. Since  $T(l, m)$  is real,  $V(-u, -v) = V^*(u, v)$ , so,  $V(u, v)$  is Hermitian and we would get two visibilities for one measurement.
5. Aperture synthesis: Sample  $V(u, v)$  at enough  $(u, v)$  points using distributed small aperture antennas to synthesize a large aperture antenna of size  $(u_{max}, v_{max})$ .

## 2.5 u-v Sampling

In a general case, one pair of antennas would give one baseline, which would in turn give two u-v samples (as  $V(u, v)$  is Hermitian) at a time. Extrapolating this,  $N$  antennas would give  $N(N-1)$  samples at a time. Refer to Figure 6 for further details.

For better imaging, we would need better coverage of the u-v plane, which can be done by observing a specific source over a period of time. Another way would be to observe the same source at multiple wavelengths, which would provide more information regarding the source.

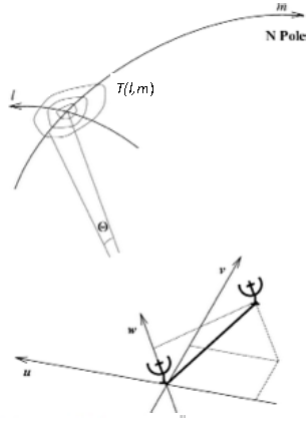


Figure 4: This shows a diagrammatic representation of the  $(u,v)$  coordinates and the  $(l,m)$  coordinates. (Source: <http://anita.edu.au/lectures/>)

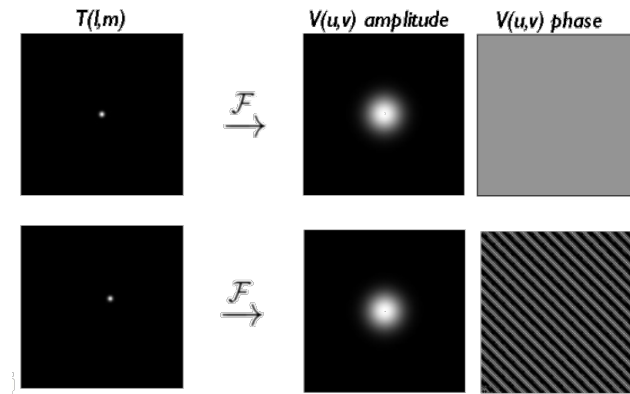


Figure 5: The first image shows the Fourier transform of a point source which is represented in amplitude and phase. Since the point source is at the center of the frame, the phase is constant. In the second figure, the point source is slightly shifted, so the phase also reflects that but because it is the same source as before, the amplitude remains the same. (Source: <http://anita.edu.au/lectures/>)

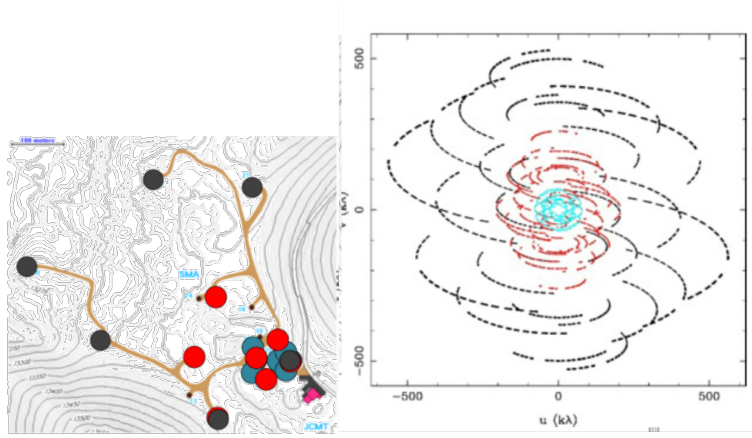


Figure 6: The first picture shows three different configurations of the SMA array represented in three different colors. They are observing a source at a Dec of +22 deg at a frequency of 345 GHz. The second image shows the total u-v coverage when observing for 8 hours. (Source: <http://anita.edu.au/lectures/>)

### 3 The Model

In this project, the system has been modeled using Python. We have used Python v2.7 wrapped with the modules of Visual Python (VPython v6.0). The graphs are plotted using both the VPython inbuilt graph function and also using Matplotlib v2.0.1. Refer to Figure 7 for further details.

Please make note that in the model, all positions and distances are measured in km, all time units are measured in seconds and all mass units are measured in kg, unless mentioned otherwise.

Earth has been assumed to be the center of the model, having a radius of  $6371\text{km}$ . In the model, the current axial tilt of the earth of  $23.4^\circ$  has been accounted for. Sun is at a distance of  $146 * 10^6\text{km}$  from the earth (center to center).

In the current version of the model, there are three satellites. If we determine that more than three satellites are required for covering the whole sky, then changes would be implemented into the model as suitable. But All the testing done till now suggests that only three satellites would be sufficiently enough to observe the whole sky with good coverage, if observed continuously for about 2 weeks. The three satellites are having orbits which are orthogonal to each other, with one of the satellite's axis superimposing the earth's rotation axis. That means one of the satellites has an equatorial orbit and the other two have perpendicular polar orbits.

All orbits have different orbital radii so as to get different baselines at different points of time. This would ensure that there is more aperture synthesis and thus, more u-v coverage. The present values of the orbital radii have been inferred upon by repeated testing.

The default distances of the satellites above earth are:-  $429\text{km}$ ,  $729\text{km}$  and  $1029\text{km}$ . So, to find the orbital radii, one need to add the earth's radius to these values respectively.

The initial velocity of the satellites has been calculated using the formula:

$$v = \sqrt{\frac{GM_e}{R}}$$

which would give the initial orbital velocities of the satellites to be:-  $7.66km/s$ ,  $7.50km/s$  and  $7.29km/s$  respectively. This formula is meant for circular orbits but actually the orbits of the satellites are elliptical in nature. This approximation was made because the orbits are nearly circular and for the sake of simpler calculations, the orbits are assumed to be circular while calculating the initial velocity.

In a similar manner, the time periods of the satellites are calculated using the formula:

$$T = \frac{2\pi R}{v}$$

which would give the time periods of the satellites assuming them to be circular orbits. But since the orbits are slightly elliptical, these values are not accurate and are just for comparison purposes. The time periods are:- *1 hour 32 minutes 56 seconds, 1 hour 34 minutes 57 seconds and 1 hour 37 minutes 36 seconds*. A similar assumption as above has been made here for the same reasons as stated above.

So, to summarize the parameters of the satellites:-

| <b>Index</b>        | <b>Distance Above Earth</b> | <b>Orbital Velocity</b> | <b>Time Period</b>    |
|---------------------|-----------------------------|-------------------------|-----------------------|
| <i>Satellite 1:</i> | 429 km                      | 7.66 km/s               | 92 minutes 56 seconds |
| <i>Satellite 2:</i> | 729 km                      | 7.50 km/s               | 94 minutes 57 seconds |
| <i>Satellite 3:</i> | 1029 km                     | 7.29 km/s               | 97 minutes 36 seconds |

During our testing, the beam angle or the field of view of each of the satellites is assumed to be 180 degrees (the whole hemisphere in front of it) but in practice, it is impossible to have such an antenna and the beam width would be much narrower. For ease of testing, such an assumption was made.

The satellites are made to revolve around the earth by equating the force of gravity due to the earth with the centrifugal force at every iteration (one second of actual time progresses at every iteration) of the program. The effect due to the sun on the satellites is also taken into account, for accurate movement. In a similar manner, the earth is made to revolve around the sun. Since the earth and the sun are the only significant celestial bodies in our case which would affect the motion of the satellites, the forces only due to them is accounted for. Forces due to the moon and other planets are ignored.

The force is equated using the formula:

$$\frac{GM_e M_s}{R^2} = \frac{M_s v^2}{R}$$

which would result in the force on the satellite. The mass of the satellite,  $M_s$  is assumed to be  $3000kg$  in our case but it is irrelevant as the mass gets canceled while equating the forces.

Please make note that the whole model is based on Cartesian coordinates  $(x, y, z)$  and all other types of coordinates are converted into Cartesian form.

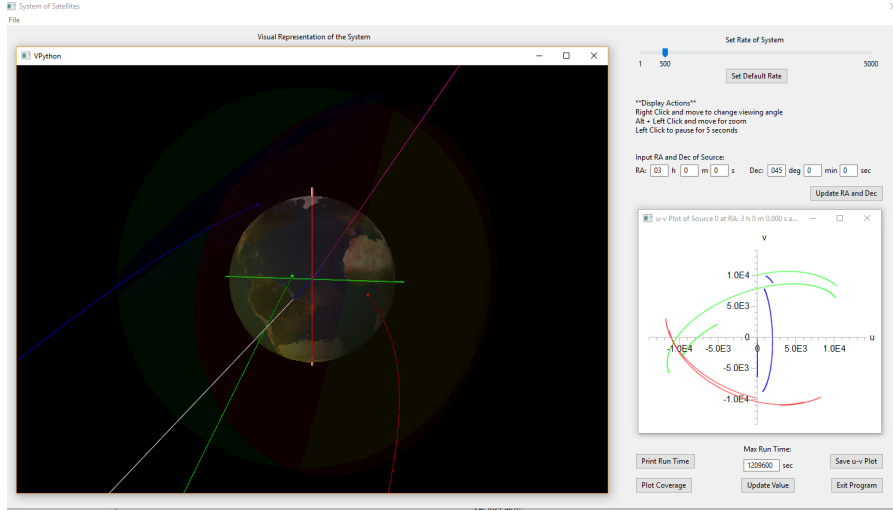


Figure 7: The screenshot of the actual model running in real time.

The line drawing of the model has been presented in Figure 8 for easier understanding. This figure has the orbits of all three satellites as well as the directions of three sources.

The user inputs the position of the source to be observed in RA and Dec (Right Ascension and Declination) which are given in hours-minutes-seconds and degrees-minutes-seconds respectively. RA varies from 0 to 24 hours whereas Dec varies from -180 to 180 degrees. In astronomy, the positions of celestial objects are usually plotted in RA and Dec, which is the standard. So, for the ease of the user, the inputs are taken in RA and Dec. These coordinates are converted into Cartesian coordinates before being processed by the program. A magenta line in the display points towards the source. In order to convert the coordinates properly, the sources are all assumed to be on a celestial sphere of a very large fixed radius (All the sources are assumed to be at the same distance from the earth. This is not a problem because in astronomy, we usually see far away objects in two dimensions only lacking any depth information and all these objects are considered to be point objects anyway.)

The formulas used to convert RA and Dec to Cartesian coordinates are:

$$a = (h_{RA} * 15 + min_{RA} * 0.25 + sec_{RA} * 0.004166) * \frac{\pi}{180}$$

$$b = (|deg_{Dec}| + \frac{min_{Dec}}{60} + \frac{sec_{Dec}}{3600}) * \frac{deg_{Dec}}{|deg_{Dec}|} * \frac{\pi}{180}$$

$$z_{source} = (d_{source} * \cos(b)) * \cos(a) - z_{earth}$$

$$x_{source} = (d_{source} * \cos(b)) * \sin(a) - x_{earth}$$

$$y_{source} = (d_{source} * \sin(b)) - y_{earth}$$



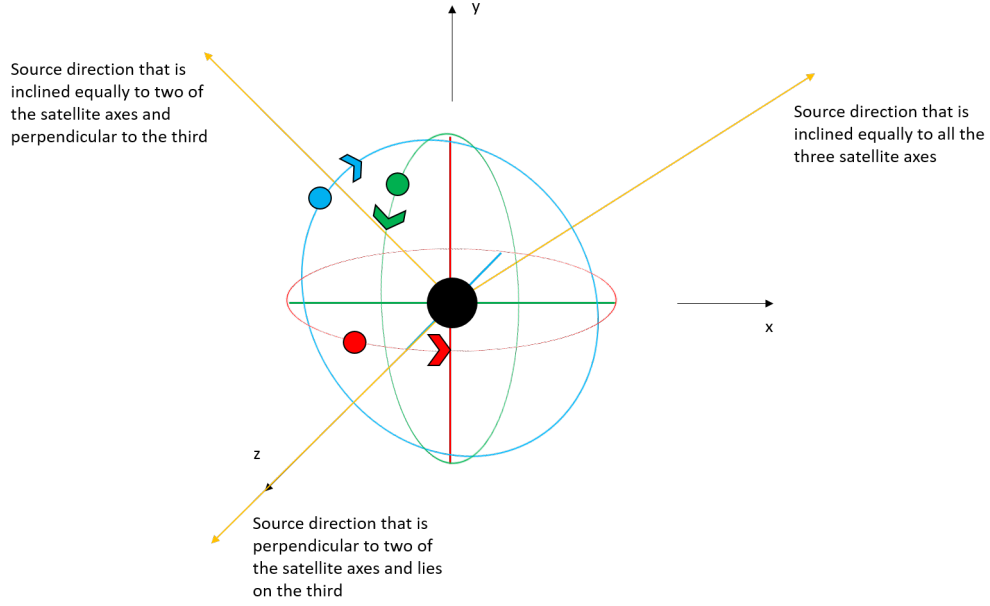


Figure 8: The line diagram of the model

where,  $(x_{source}, y_{source}, z_{source})$  are the Cartesian coordinates of the source as converted from the RA and Dec coordinates.  $d_{source}$  is assumed to be  $300 * 10^6 km$ .  $(x_{earth}, y_{earth}, z_{earth})$  are the Cartesian coordinates of the earth in its current position.

The user is also provided with the option of defining the frame rate of the system, which is initially set to 500 frames/second, with each frame displaying one second of the actual time. The user can vary the frame rate from 1 to 5000 frames/second. Since the code is very compute intensive, so it is unlikely to achieve a frame rate greater than 5000, and hence is not provided.

Initially, the source is set to a RA of  $3hrs$  and a Dec of  $45^\circ$ , which is a source that is at equal angles from all three satellite orbit axes. This would give us an idea of the coverage with the current configuration. The source position can be updated mid program to as many times as required.

Currently, the program is set to stop automatically after running for 1209600 seconds or 2 weeks. But this maximum runtime can be updated mid program. If the max runtime is set lower than the current runtime, the program will terminate immediately.

The visual display has its own set of commands. Right click with the mouse on the display and move to change the viewing angle. Use ALT + Left click and move to zoom in and out of the display. Left click once to pause for 5 seconds and Left click  $n$  times to pause for  $n * 5$  seconds.

The visual display is compute intensive and may cause lag. So it can be closed separately in order to improve the system performance. Once closed, the display cannot be brought back and the above mentioned display commands would also cease to work.

There are several click-able buttons on the window, each serving a different purpose. The functions in the top right are self explanatory. For those in the bottom right, the “Print Run Time” button prints the time for which the system has been running in seconds. The “Save u-v Plot” button copies all the existing u-v plot data into an array which would then be saved into a PDF file using Matplotlib. Do note that when the source is updated, the arrays are cleared so as to avoid overlap of plotting data. The “Exit Program” button plots the u-v plot and then exits the program. The “Plot coverage” button does a similar thing as the “Save u-v Plot” function but instead of saving the plot exactly as it is, it grids the u-v map into a grid of cell size  $nkm \times nkm$  (where  $n$  is 100, by default) and if even one point falls inside a cell, the cell is considered as filled. This is plotted for each satellite pair individually and then for all three satellites and then the percentage coverage is also calculated.

All the relevant details of the system are printed in the “Python Shell” window for user reference. Relevant details include the name of the file being run, the satellite heights above earth surface, their initial velocities and time periods, the source positions, the number of sources observed, the time for which the system has been running, failures and errors if any, and others.

The u-v coverage is measured, as a standard, unless mentioned otherwise, for a time of two weeks. In the u-v coverage plot, red represents the baseline for satellite 1 and 2, blue represents the baseline for satellite 2 and 3, green represents the baseline for satellite 3 and 1. Refer to Figure 9 for further details.

The u and v values are currently measured in km and not in wavelengths (which is the standard). In the current model, the Hermitian symmetric nature of u-v is not reflected, and only one u-v point is plotted per baseline per time. This is done to avoid cluttering the graph with too many points. The final u-v plot would be created once the testing phase is over. Another plot is shown in Figure 10, which is the same as Figure 9 but it takes into account the Hermitian symmetric nature of the function and shows two points plotted per baseline. In all the further shown u-v plots in this report, the Hermitian symmetric nature has been ignored, unless mentioned otherwise.

## 4 Procedure

The procedure for plotting the u-v coverage is explained here. Let us assume that the source is pointing towards a particular direction  $\vec{d}$ . Then, we define two unit vectors  $\hat{u}$  and  $\hat{v}$  such that they are perpendicular to  $\vec{d}$  and are also perpendicular to each other. These vectors form the u-v axes for the u-v plane. Do note that these axes change with the change in direction of the source, for obvious reasons.

Next, let us assume that at a particular instant of time, any two satellites are at positions  $(x_a, y_a, z_a)$  and  $(x_b, y_b, z_b)$  in space respectively, and the position of the center of the earth is  $(x, y, z)$ . We also assume that the source is in the field of view (FoV) of both these satellites at this particular instant. The FoV of the satellites, currently, is the whole 180 degrees of the hemisphere that the satellite is facing. We check whether the source falls in the satellites field of view by doing a dot product between  $\vec{d}$  and  $(\vec{x}_b - \vec{x}, \vec{y}_b - \vec{y}, \vec{z}_b - \vec{z})$ . If this dot product comes out to be positive, then we can confirm that the source is within the FoV of the satellite.

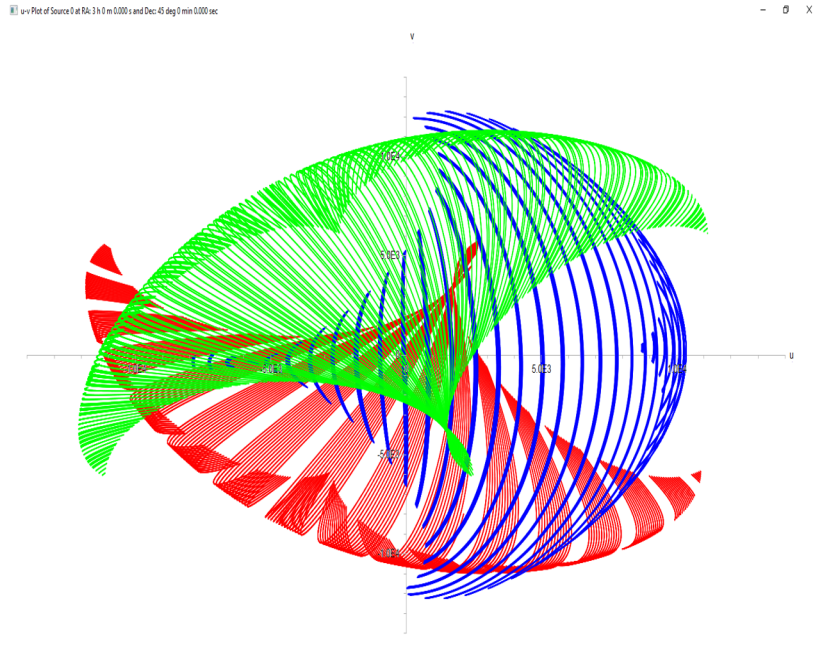


Figure 9: The u-v coverage plot for 2 weeks for a source that is equidistant from all the satellite orbits. Please note that this does not plot the Hermitian symmetric nature of u-v.

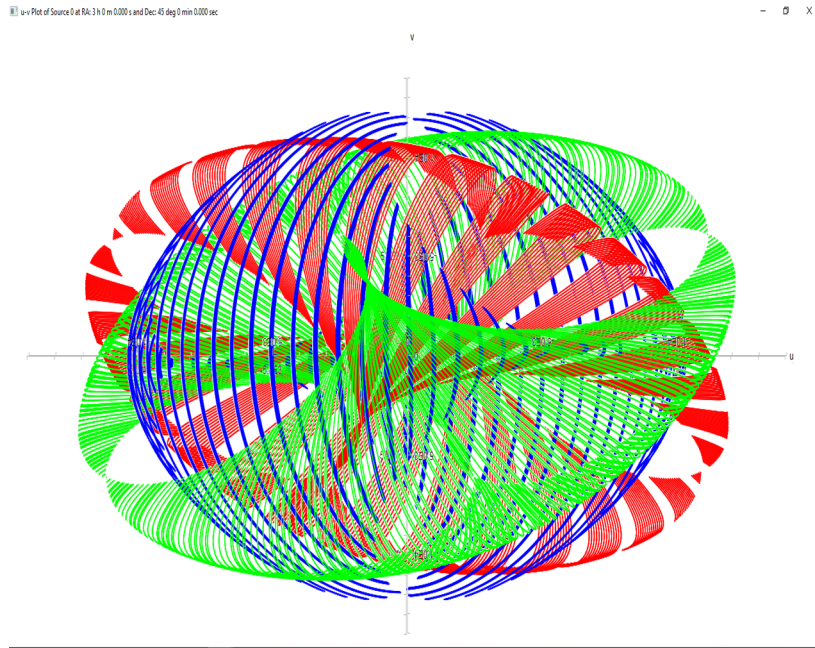


Figure 10: The u-v coverage plot for 2 weeks for a source that is equidistant from all the satellite orbits. This plot shows the Hermitian symmetric nature of u-v.

Then, we define a vector pointing towards the second satellite from the first, which would be defined by  $(\overline{x_b - x_a}, \overline{y_b - y_a}, \overline{z_b - z_a})$ . Then, we do a dot product between  $(\overline{x_b - x_a}, \overline{y_b - y_a}, \overline{z_b - z_a})$  and  $\hat{u}$  which would give the value of  $u$ . In a similar manner, we get  $v$  by doing a cross product between  $(\overline{x_b - x_a}, \overline{y_b - y_a}, \overline{z_b - z_a})$  and  $\hat{v}$ .

Thus, for a particular instant of time, we get the value of  $(u, v)$ . And over a duration of time, we are able to develop the u-v plot for a baseline between any two satellites, when the source is in their field of view.

## 5 Results

Presently, the simulation of the model is complete, and we are also at the end of the testing phase, where different parameters are modified and tested so as to obtain the best coverage of the sky. The current results include:-

1. The initial model plotted  $(u, v)$  at the intervals of every second, which may be unnecessary. This means that there was an average of 10 km spacing between any two  $(u, v)$  points. Since this value is much larger than the observed wavelength (which is about 30 m), it would not matter if we plot the  $(u, v)$  at greater time intervals, as long as the deviation of the plot is not significant.

Tests were run for different plotting intervals like 5 seconds, 10 seconds and 20 seconds. In each case, the average spacing was 53 km, 95 km and 215 km respectively. Then a histogram was plotted between a few of the sample points to calculate the mean deviation from the 1 second plotting and the values were respectively found to be 0.10 km, 0.14 km and 0.55 km.

So, for the most efficient process, the plotting was decided to be done at intervals of every 10 seconds.

2. Test for the ultimate coverage of the system, which was done by choosing three source directions. One, which is at equal angles from all three satellite orbit axes; the second which is perpendicular to one of the satellite orbit axis and is at equal angles from the other two; and the third which is perpendicular to any two satellite orbit axes and lies on the third. All the coverages were plotted for 2 weeks. Refer to Figures 11, 12 and 13 for further details.

This test proved that the system of three satellites has sufficiently enough coverage for all of the source positions anywhere in the sky. Do note that the beam sensitivity and beam angle considerations are ignored in this and all of the graphs shown henceforth. Those considerations, even though would make the system less effective, are necessary to be considered for the practical case and shall be accounted for in future developments.

3. Test for sensitivity of the system by changing the orbital period of any one satellite, and then repeating the procedure with the other two satellites. All the plotting was done for 1 week. Since the orbital period is related to its distance above the earth, so we would be varying the distance and those values shall be mentioned here henceforth. These values can be converted into time period by using the specified formula. All the tests

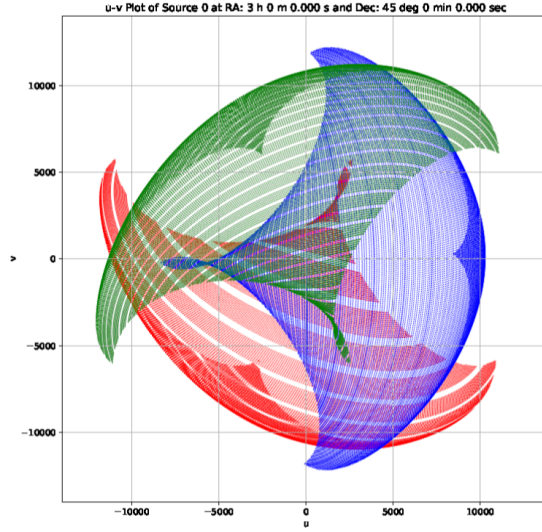


Figure 11: The u-v coverage plot for 2 weeks for a source direction that is inclined equally to all the three satellite orbit axes. The RA and Dec of the source are 3 h 0 min 0 sec and 45 deg 0 min 0 sec respectively. This plot does not show the Hermitian symmetric nature of u-v.

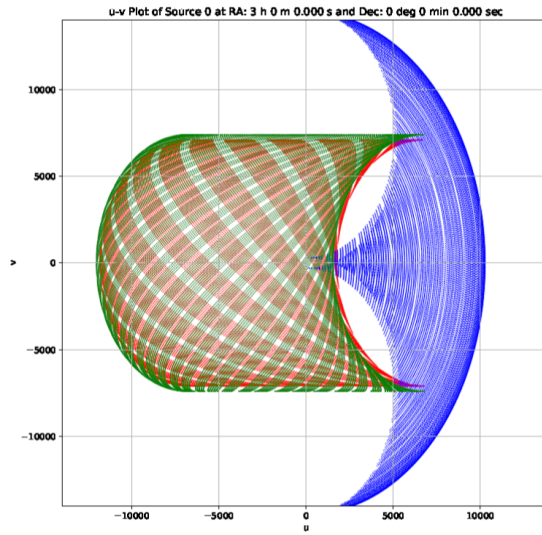


Figure 12: The u-v coverage plot for 2 weeks for a source direction that is inclined equally to two of the satellite orbit axes and perpendicular to the third. The RA and Dec of the source are 3 h 0 min 0 sec and 0 deg 0 min 0 sec respectively. This plot does not show the Hermitian symmetric nature of u-v.

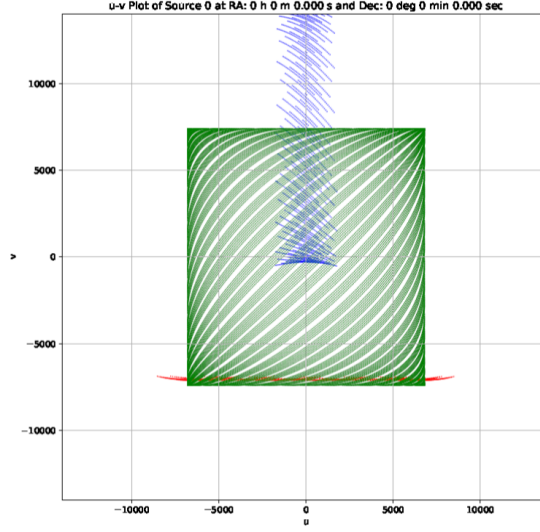


Figure 13: The u-v coverage plot for 2 weeks for a source direction that is perpendicular to two of the satellite orbit axes and lies on the third. The RA and Dec of the source are 0 h 0 min 0 sec and 0 deg 0 min 0 sec respectively. This plot does not show the Hermitian symmetric nature of u-v.

have been done for the source direction that is inclined equally to all the three satellite orbit axes. Only a few of the test plots have been included. Refer to Figure 14 for further details and results.

4. Test by using different combinations of their initial phases, so as to find the best coverage for a source which is at equal angles from all three satellite orbit axes. All the plotting was done for 2 weeks. Only a few of the test plots have been included. Refer to Figure 15 for further details and results.

Please note that while calculating the initial phase of the satellites (It is 0 degrees by default), the satellite orbits are again assumed to be circular.

5. Test for quantitatively measuring the coverage for all the three ultimate source directions. This was done by coarsely gridding the  $(u, v)$  plane, that is, dividing the plane into a grid of a certain cell size (100 km) and then considering the cell as filled if one or more  $(u, v)$  points fall within that cell. This grid plot would show the coverage and then percentage coverage was calculated by considering the whole  $(u, v)$  plane as a circle of radius 15000 km (maximum baseline length). Do note that if the Hermitian symmetric nature is considered, the coverages would increase significantly.

For source direction that is inclined equally to all the three satellite orbit axes, the percentage coverages are:-

- (a) For baseline due to satellites 1 and 2: 42.6795%
- (b) For baseline due to satellites 2 and 3: 43.3911%

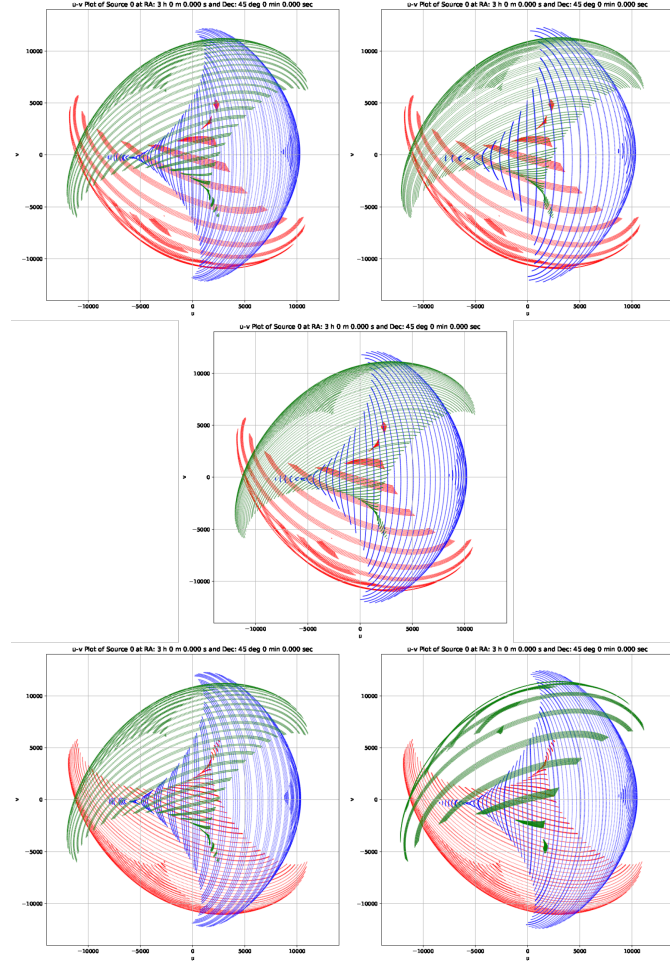


Figure 14: The u-v coverage plot for 1 week for a source direction that is inclined equally to all the three satellite orbit axes. This plot does not show the Hermitian symmetric nature of u-v. In usual order, the plots have the following satellite configurations:- Satellite 1: 429 km, Satellite 2: 729 km, Satellite 3: 1029 km — Satellite 1: 429 km, Satellite 2: 729 km, Satellite 3: 1129 km — Satellite 1: 429 km, Satellite 2: 729 km, Satellite 3: 929 km — Satellite 1: 429 km, Satellite 2: 829 km, Satellite 3: 1029 km — Satellite 1: 429 km, Satellite 2: 829 km, Satellite 3: 1229 km — From these plots we can conclude that for the fourth configuration, the coverage seems to be the most uniform and covers a larger area.



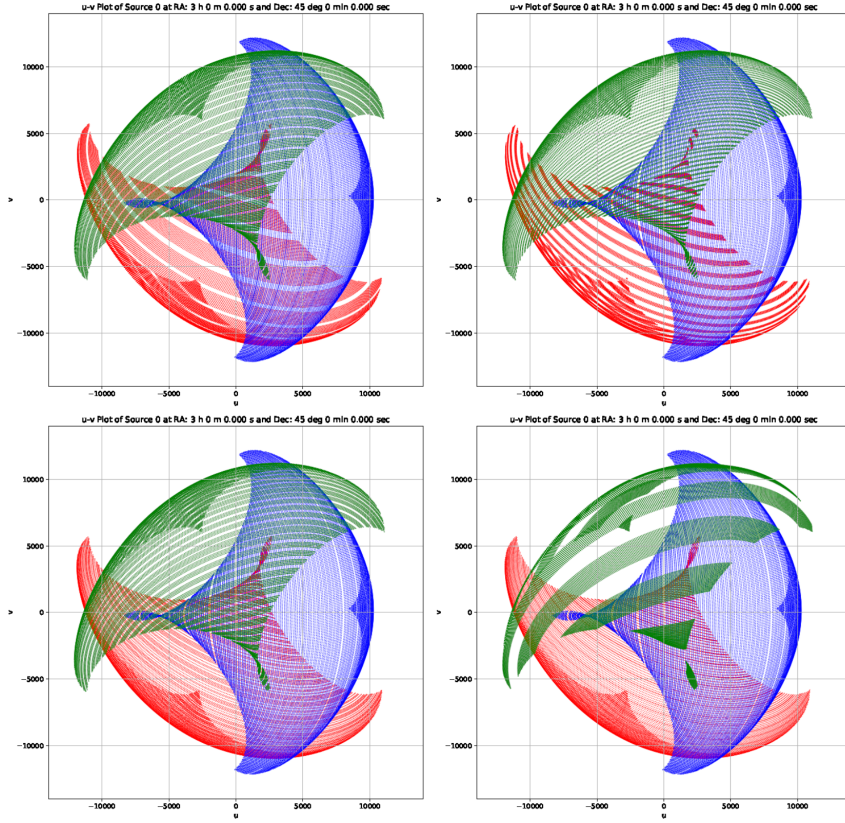


Figure 15: The u-v coverage plot for 2 weeks for a source direction that is inclined equally to all the three satellite orbit axes. This plot does not show the Hermitian symmetric nature of u-v. In usual order, the plots have the following phase variations for Satellite 1:- 0 degrees, 10 degrees, 30 degrees, 45 degrees — From these plots we can conclude that for the configuration with 30 degrees phase, the coverage seems to be the most uniform and covers a larger area.

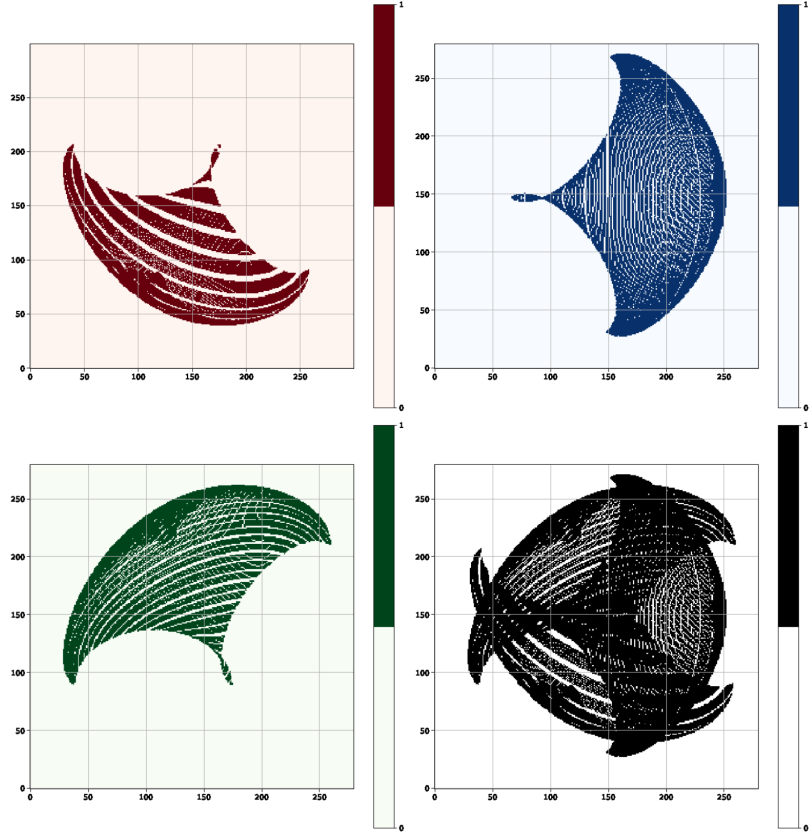


Figure 16: The u-v coarse gridding plots for 2 weeks for a source direction that is inclined equally to all the three satellite orbit axes. This plot does not show the Hermitian symmetric nature of u-v. In usual order, the plots represent the coverage due to baseline of satellites 1 and 2, 2 and 3, 3 and 1, and finally, due to the baselines of all the three satellites.

- (c) For baseline due to satellites 3 and 1: 42.9596%
- (d) For baselines due to all three satellites: 71.2004%

Refer to Figure 16 for further details and results.

For source direction that is inclined equally to two of the satellite orbit axes and perpendicular to the third, the percentage coverages are:-

- (a) For baseline due to satellites 1 and 2: 42.0485%
- (b) For baseline due to satellites 2 and 3: 42.3625%
- (c) For baseline due to satellites 3 and 1: 43.1109%
- (d) For baselines due to all three satellites: 65.8711%

Refer to Figure 17 for further details and results.

For source direction that is perpendicular to two of the satellite orbit axes and lies on the third, the percentage coverages are:-

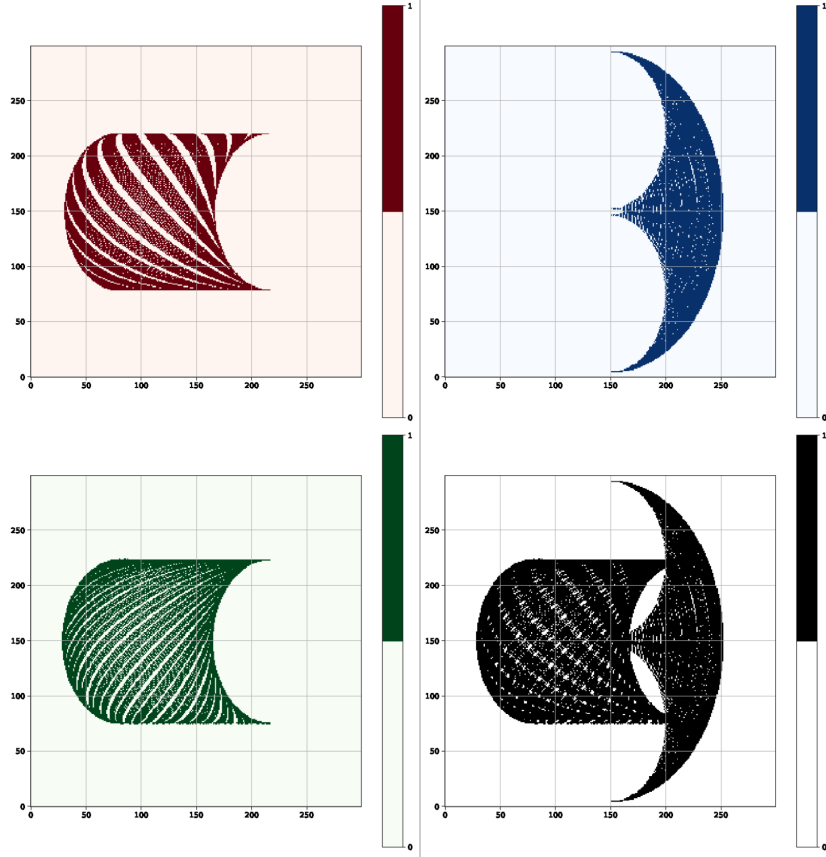


Figure 17: The u-v coarse gridding plots for 2 weeks for a source direction that is inclined equally to two of the satellite orbit axes and perpendicular to the third. This plot does not show the Hermitian symmetric nature of u-v. In usual order, the plots represent the coverage due to baseline of satellites 1 and 2, 2 and 3, 3 and 1, and finally, due to the baselines of all the three satellites.

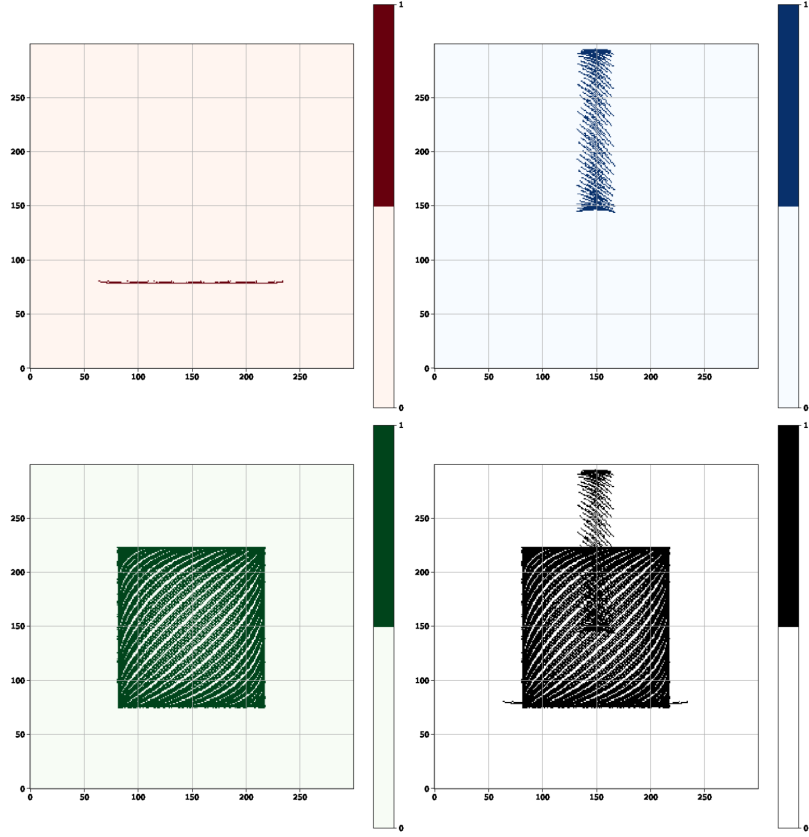


Figure 18: The u-v coarse gridding plots for 2 weeks for a source direction that is perpendicular to two of the satellite orbit axes and lies on the third. This plot does not show the Hermitian symmetric nature of u-v. In usual order, the plots represent the coverage due to baseline of satellites 1 and 2, 2 and 3, 3 and 1, and finally, due to the baselines of all the three satellites.

- (a) For baseline due to satellites 1 and 2: 3.6372%
- (b) For baseline due to satellites 2 and 3: 3.5071%
- (c) For baseline due to satellites 3 and 1: 43.1859%
- (d) For baselines due to all three satellites: 46.3818%

Refer to Figure 18 for further details and results.

## 6 Ongoing Work and Future Investigations

While the major work has been finished, the model still does not reflect to the practical considerations of beam width, which shall be addressed in the near future. Then, in order to see if the model is working as expected in real life conditions, it's outcomes would be compared to other accepted outcomes. Finally, the science goals of the system would be decided.

The ongoing work includes:-

1. We would undertake the calculations of finding the beam efficiency and the main beam width.

Whereas, the future investigations would include:-

1. The sky maps for 35 MHz and 400 MHz would be obtained so as to compare the generated sky map for our system and perform a detailed study.
2. Conduct a literature survey about ionized hydrogen regions and general electron density of the interstellar medium for the second part of the project.

## 7 Relevant Work in this Field

There have been several instances of research which included similar setups of space based interferometers or earth-space based interferometers in the past for various radio and other astronomical studies. Some of the popular ones are mentioned here:-

1. NASA's Space Interferometry Mission (SIM Lite) was proposed to hunt for earth sized planets in habitable zones of nearby stars. It consisted of a 2-element Michelson optical interferometer with a baseline of 6 m. Unfortunately, it was canceled in 2010. (Source: <https://science.nasa.gov/missions/sim>)
2. ESA's Laser Interferometer Space Antenna (LISA) has been proposed to detect and measure gravitational waves with much greater accuracy, as compared to LIGO. It consists of a 3-element laser interferometer in a heliocentric orbit with 2.5 million km baseline. It is set to launch set in 2034. (Source: <https://www.elisascience.org/>)
3. Sun Radio Interferometer Spcae Experiment (SunRISE) has been proposed to track particle acceleration and transport in the inner heliosphere, especially the radio emissions due to coronal mass ejections from the sun. It consists of a 6-element interferometer with a baseline fo 10 km. It was proposed in 2017. (Source: <http://ieeexplore.ieee.org/document/7943789/>)
4. NASA's Space Very Long Baseline Interferometry was proposed to study high resolution radio images using a combination of space and earth based antennas, providing a baseline greater than 2.5 times the diameter of the earth. It was launched in 1997. (Source: <https://www.jpl.nasa.gov/missions/space-very-long-baseline-interferometry-space-vlbi/>)
5. Roscosmos's RadioAstron, an earth-space based radio interferometer, having a 10 m dish in a highly elliptical orbit with a baseline of 350,000 km. It operates in four frequency bands, 1.3, 6, 18 and 92 cm. It was launched in 2011. (Source: <https://arxiv.org/pdf/1303.5200>)

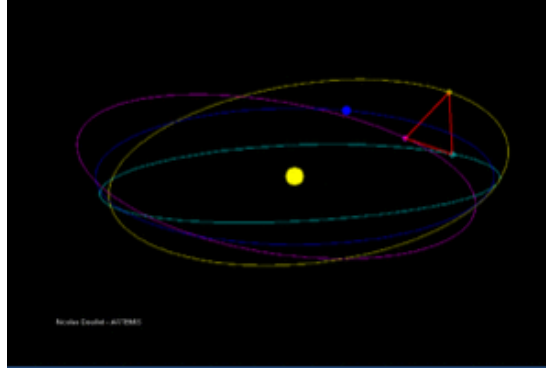


Figure 19: An artist's perception of e-LISA where the three interferometers form a triangular baseline and are following behind the earth.

6. ISAS's MUSES-B program, an earth-space based radio interferometer, having a 8 m dish in a medium earth orbit with a baseline of 20,000 km. It was launched in 1997. (*Source: <http://space.skyrocket.de/doc-sdat/muses-b.htm>*)

From all these projects, we can know that there has never been a fully space based radio interferometer array, which is specifically designed to detect radio waves having wavelengths greater than 30 m. Hence, our model would be a state of the art and would lead to newer discoveries.

Presently, the Indian Space Research Organization (ISRO) is calling for proposals from enthusiastic research groups towards making a space based instrument to radio astronomical observations. Interested people can submit their proposals to ISRO and if short listed, they would proceed to build the proposed instrument. This could be a great prospective opportunity concerning our project.

## 8 Major Contributions

Our project would be doing the following major contributions to the research in radio astronomy:-

1. This would be the world's first fully space based VLBI system with a minimum of three interferometers, that would be made in India.
2. This system would provide huge baselines of over 13,000 km, much greater than earth based interferometers, thus, providing with much better resolutions.
3. Our setup can perform observations for frequencies below 30 MHz, which are not possible by earth based radio telescopes.
4. The telescopes we would be using would have wide band antennas which would provide excellent coverage of the sky, at the cost of reduced sensitivity.

5. The u-v coverage would be much greater and faster than any earth based or earth-space based interferometers.
6. The data processing can be done on board, reducing the requirements of high data transfers and avoiding loss of data and easier to dump final data.

## 9 Conclusion

All the required concepts involved in the simulation of the minimal space interferometer configuration such as low frequency observations, low earth orbit satellites, very long baseline interferometry, visibility function, u-v sampling, etc. to name a few, were successfully studied during the initial period of the project.

The model was successfully simulated using Python programming with Python v2.7 and wrapped with the modules of VPython v6.0. The model consists of a three satellite interferometer, with low earth orbit satellites having different orbital periods and orthogonal orbits.

A system for user to input the source coordinates in RA and Dec was designed and the coordinates were further converted into (x,y,z) to be processed by the program. A pointing mechanism was also designed to give user the idea of the direction of the source.

The various properties of the model as well as the assumptions and approximations done in the model are clearly mentioned. Also, the procedure for plotting the u-v coverage has been elaborated to give a clear view of the process to the reader.

The u-v coverage of different sources was measured and the various parameters in the system were varied so as to get maximum u-v coverage for the entire sky.

In future work, different observations would be combined to obtain a complete images of the sky at a range of low radio frequencies, which would then be compared with their counterparts at higher frequencies to study the spectral evolution, and possible new features emerging at the low radio frequencies.

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